

Mohamed Abdelrahman

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EDUCATION

University of Houston

Bachelor of Science in Computer Science

- GPA: 3.28 / 4.00

Aug. 2023 – May 2026

Houston, TX

EXPERIENCE

Code[Coogs] Competitions Director

Jan 2025 – May 2025

University Of Houston

Houston, TX

- Led 9 competitive programming events using C++ to teach data structures, graph algorithms, and complexity analysis to 30+ students
- Organized and hosted algorithmic contests on Kattis, curating problem sets aligned with LeetCode-style technical interview topics
- Hosted interactive workshops with live coding, peer discussions, and real-time feedback to improve student engagement
- Collaborated with student leaders and faculty to expand outreach and iterate on event formats based on collected feedback

Coding Expert - AI Trainer (Contract)

Aug 2023 – Jan 2025

Outlier.ai

Remote

- Developed solutions for fine-tuning code generation models using Python and C++, improving quality by 40%
- Created prompts to produce model deviations that improve model training using prompt engineering techniques
- Annotated LLM output for over 400 responses and rated its quality based on a rigorous rubric
- Reviewed submissions for other contributors and provided feedback, improving their task quality by 30%

Software Engineering Intern

Aug. 2022 – Jan 2023

bld.ai

Remote

- Collaborated with four interns to develop a full-stack application using React, Django, and PostgreSQL
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month, following a two-week sprint cycle
- Followed Agile Practices including sprint planning, daily stand-ups, and retrospectives to optimize team workflow by 50%

PROJECTS

C++ Ordered Map Library | C++17, CMake, LCOV, Valgrind, Catch2, GitHub Actions

- Implemented an ordered map data structure by building a doubly linked list from scratch using smart pointers and custom iterators, then wrapping it with an unordered map for O(1) key lookups while maintaining insertion order
- Achieved 97.3% code coverage across 2,246 test assertions using LCOV and Valgrind for comprehensive testing
- Designed STL-compatible interface with iterator support, exception safety, and template-based architecture using modern C++17 features
- Established CI/CD pipeline with automated testing and memory leak detection using GitHub Actions and CMake
- Verified zero memory leaks through Valgrind analysis and documented comprehensive API with usage examples

Nota | Flutter, Dart, Bloc, SQFlite, Quill, Material 3

- Developed a rich-text note-taking app for Android with offline storage using SQL queries in SQFlite
- Integrated Flutter Quill editor to support formatting such as bold, italics, headers, links, and lists
- Implemented labeling and filtering features for efficient note organization and retrieval
- Managed UI and business logic using Bloc state management for clean separation of concerns
- Designed a responsive UI with Material 3 components and support for light/dark theme switching

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart

Frameworks: Django, React, NextJS, Flutter

Developer Tools: Git, Docker, Postman, VS Code, CLion, Visual Studio, PyCharm, IntelliJ, DBeaver

Libraries: unittest, React Router, Bloc, Flutter Quill, SQFlite, Axios